SELF-EVOLUTION SHOULD NOT DEPEND ON HOW SKILLS ARE SPLIT: AN EXACT CERTIFICATE FOR SKILL-TAXONOMY REPRESENTATION INVARIANCE

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ABSTRACT

Self-evolving language-model agents increasingly treat skill identities as control objects: a sampled skill can determine the next training task, a fresh ID can receive retrieval priority, and feedback can be written back to the selected package. We ask whether such control should change when semantic capabilities are merely split or cloned across package identities. We formulate *Skill-Taxonomy Representation Invariance* (STRI). For exact refinements, we prove that invariance is equivalent to factoring aggregate semantic-class mass through the quotient; any strictly positive identity-local score followed by normalization is therefore clone-sensitive. Released Skill-SP instantiates this failure mode at initialization: 15 packages receive probability $1/15$ each, double-supported Level-1 rows receive twice the additive eligibility opportunity of singleton rows, and a counterfactual same-state sampler-input clone changes its prompt-mixture by TV $= 0.0583$; quotient-conserved mass restores it exactly. For partial overlap, define $R^*(A; q) = \min\{t : q \leq Aw \leq tq,\, w, t \geq 0\}$ for support matrix A and representation-independent target $q > 0$. It equals one iff q lies in the package support cone. On Skill-SP Level-1, neutral $R^* = 2$ on full, calibration, and tool-disjoint heldout matrices and survives exhaustive one-cell stress tests; Level-3 and the 127/128-overlap logical domain are equalizable. In one archived AutoSkill P19 substrate, splitting one skill into four exact-semantic IDs deterministically crowds out a post-checkout skill and removes a mechanically defined destructive behavior (6/6 original versus 0/6 split), while ID-placebo and quotient controls restore it (3/3 each; Fisher $p = 0.00108$). Under the split representation, adding that post-checkout skill back restores the behavior in 3/3 fresh runs while a matched cleanup add-back yields 0/3 (exact one-sided Fisher $= 1/20$), isolating the mediator within this substrate. Thus representation can alter executed behavior on this bounded substrate without semantic change, whereas overlap prevalence alone does not determine the cone residual. A semantic-first construction marks the action-basis escape when package-only control cannot realize q; it is constructive, not a validated repair.

1 INTRODUCTION

Many self-evolving agents adapt through external state—memories, skills, workflows, tools, prompts, retrieval policies, or harness code—rather than only through foundation-model weights. Skills are especially attractive because a skill can package reusable procedure, metadata, examples, applicability information, and verification logic. Recent systems make the skill identity itself a control variable: Skill-SP (Huang et al., 2026) samples a skill before generating a training task, while SkillRL (Xia et al., 2026) gives dynamically created skill IDs priority in a fixed retrieval budget.

This architecture makes the *representation* of the skill library consequential. Consider two taxonomies that expose the same executable semantic capability. One stores a primitive once; another stores an exact clone under a fresh ID. One represents generic and specific capabilities separately; another groups unchanged primitives behind a macro package. If no new primitive behavior or support region is added, should the future curriculum, retrieval exposure, or credit allocation change solely because package bookkeeping changed?

Existing work on semantic skill clones, redundancy governance, skill safety, adaptive routing, compression, and structured bandits establishes that skill quality, redundancy, and routing matter. These lines do not define a representation-invariance property for an endogenous evolution controller. We therefore ask: *when can package-indexed control be made invariant to semantics-preserving changes in the skill taxonomy using the same frozen support information?*

We approach the question asset-first. Rather than proposing a normalization method and manufacturing a benchmark, we inspect released control surfaces, reduce the phenomenon against strongest same-information baselines, and identify a support geometry that separates positive and negative regimes.

Released Skill-SP. On API-Bank Level-1, 314 released tasks are covered by at least one released validator and 183 have multiple memberships. The release's name+description Jaccard duplicate filter at threshold 0.33 misses all five observed specific–generic support-overlap pairs. Minimum-cardinality whole-package pruning can reduce the active library from 10 to 7 while preserving exact support, but 71 overlapping tasks remain. The residual is therefore not merely an exact-clone artifact.

Released SkillRL. In released template-mode retrieval, dynamic general skills are included before the remaining fixed `top_k=6` budget is filled with static skills. Adding one exact-content fresh-ID copy at a time admits all 12 tested clones. Across 223 released ALFWorld task descriptions, 11/12 clone targets change the unique semantic retrieval set and 5/12 reduce the number of unique general-skill contents returned.

These systems establish representation sensitivity, but not its boundary. The logical domain provides a direct counterexample to overlap-as-sufficient-explanation: 127/128 tasks overlap while the neutral target remains exactly realizable. Thus overlap count alone cannot determine the STRI residual.

Contributions.

1. **STRI problem and released evidence.** We formulate representation invariance for skill-indexed self-evolution and document identity sensitivity in two independent released systems.
2. **Controller characterization.** We prove exact-refinement STRI is equivalent to quotient factorization of semantic-class mass, and that positive identity-local score normalization is necessarily clone-sensitive; when the exact clone class is known, quotient-conserved mass exactly restores this pure multiplicity nuisance at the allocation level.
3. **Geometry certificate and design boundary.** We derive $R^*(A; q)$, its exact dual, robust/feasible-set variants, and a semantic-first construction that realizes any covered target once the controller is allowed to choose semantic intent before implementation. Thus $R^* > 1$ identifies insufficiency of the package-first action basis rather than a universal impossibility.
4. **Released and dynamic evidence.** Skill-SP exhibits the controller failure and a Level-1 residual with Level-3/logical equalizable boundaries; AutoSkill provides a controlled one-substrate representation-to-retrieval-to-behavior chain with ID-placebo, quotient restoration, and matched mediator add-back; SkillRL provides an independent identity-sensitivity boundary.

2 From Package Identity to Semantic Exposure

2.1 Frozen support matrix

Let $X = \{x_1, \ldots, x_n\}$ be a finite set of semantic contexts or tasks with independently defined first-party support truth, and let $S = \{s_1, \ldots, s_m\}$ be the current skill packages. Define the binary incidence matrix

$$A_{ij} = \mathbf{1}\{s_j \text{ supports } x_i\}.$$

The support definition is frozen before evaluating STRI. In the tool-call study, support comes from released Skill-SP validators. In the logical-reasoning study, tasks are generated by released compiler

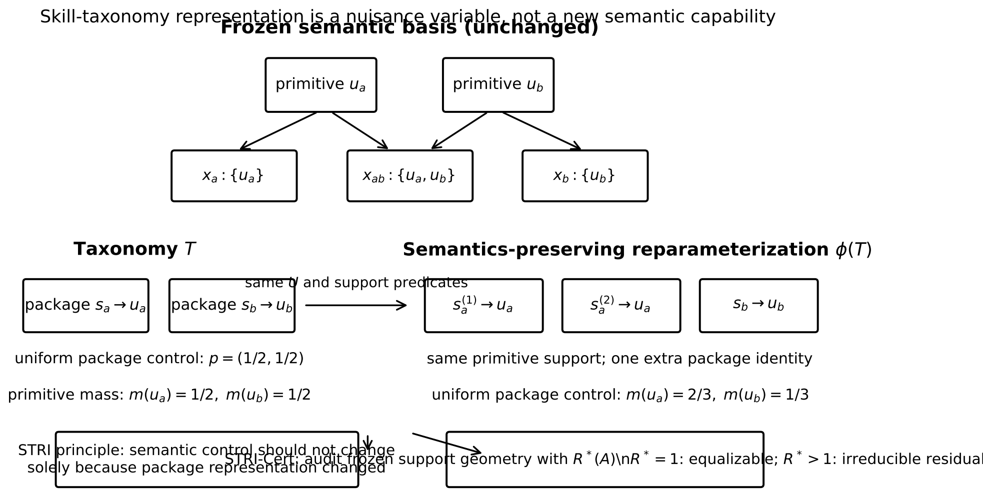

Figure 1: STRI separates semantic capability from package representation. The conceptual identity-split example leaves the primitive basis and support predicates unchanged, yet a package-indexed controller can change primitive mass solely because one primitive has two package identities. Exact clones motivate the nuisance channel; the decisive released residual in Sec. 4 survives clone reasoning and is characterized by non-equalizable partial-overlap support geometry.

specifications, checked by the author's fixed CSP/contract code, and cross-skill support is determined by the released compiler-alignment predicate.

We study package-only pre-context control. At a frozen state, the controller chooses nonnegative package mass $w \in \mathbb{R}^m_{\geq 0}$ before the semantic task to be generated exists. The additive eligibility exposure of context x_i is $e_i = (Aw)_i$. This is a control-opportunity quantity, not a task probability and not utility. Let $q \in \mathbb{R}^n_{>0}$ denote a *representation-independent semantic target*: it may encode an externally justified nonuniform priority, but it may not be changed merely because packages are renamed, cloned, split, or regrouped. When no such pre-context semantic preference is specified, we use the neutral target $q = \mathbf{1}$.

2.2 Same-information pre-task controllers reduce to package mass

In released Skill-SP, skill selection occurs before task generation. At this decision point the action is one current package identity. For any fixed pre-task state z, every randomized same-information rule—including a bandit, MLP, or neural router—induces a categorical distribution over packages, hence one nonnegative mass vector $w(z)$. The strongest same-information baseline is therefore arbitrary nonnegative package mass with access to the complete frozen support matrix.

This reduction is pointwise in z. It does *not* claim that a longitudinal adaptive controller is globally equivalent to one static vector, and it does not allow a baseline to condition on the future generated task when the reviewed controller acts before that task exists.

2.3 Exact-refinement STRI factors through a quotient

Let $\mathcal{R}$ be a package representation and let $Q(\mathcal{R})$ collapse exact semantic clones—packages with identical semantics, support, and controller-relevant state—into classes. Let $G_{\mathcal{R}}$ sum a package-mass vector over these classes. For a fixed pre-context state z, define *exact-refinement STRI* by

$$G_{\mathcal{R}} p_{\mathcal{R}}(z) = G_{\mathcal{R}'} p_{\mathcal{R}'}(z) \quad \text{whenever } Q(\mathcal{R}) = Q(\mathcal{R}').$$

Proposition 1 (quotient-factorization characterization). A controller family satisfies exact-refinement STRI iff there exists a quotient-level map $\bar{p}$ such that $G_{\mathcal{R}} p_{\mathcal{R}}(z) = \bar{p}(Q(\mathcal{R}), z)$ for every representation and state.

Proof. The forward direction defines $\bar{p}$ using any representative of a quotient class; STRI makes this definition representative-independent. The reverse direction is immediate because equal quotients give equal $\bar{p}$. $\square$

Corollary 1 (identity-local normalization is clone-sensitive). Suppose $p_j = g(h_j, z)/\sum_k g(h_k, z)$ with $g > 0$ and an exact clone receives the same local state h_j. Writing its score as $a > 0$ and all other scores as $B > 0$, the semantic-class mass changes from $a/(a + B)$ to $2a/(2a + B)$, an increase of $aB/((a + B)(2a + B)) > 0$. Repeated clones drive class mass toward one, so exact-refinement invariance requires class-aware conservation or another quotient-factorized rule.

Proposition 2 (semantic-first construction handles arbitrary overlap). Normalize a target ray $q > 0$ to $\pi = q/(\mathbf{1}^\top q)$. If every target-positive semantic unit i has at least one supporting package, choose any row-stochastic kernel K with $K_{ij} = 0$ whenever $A_{ij} = 0$, then sample $i \sim \pi$ before sampling implementation $j \sim K_i$. The semantic marginal is exactly π for arbitrary overlap, and any clone/split can redistribute K_i within the corresponding implementation class without changing that marginal.

Proof. The joint mass is $J_{ij} = \pi_i K_{ij}$, so $\sum_j J_{ij} = \pi_i \sum_j K_{ij} = \pi_i$; support-constrained row-stochastic K exists exactly when every positive-target row is covered. $\square$ This changes the action interface: it requires representation-independent semantic intent and conditional implementation. It is a constructive action-basis witness for escaping package-first infeasibility, not a same-information baseline or validated downstream method.

3 Target Realizability and STRI-Cert

3.1 Target-conditioned finite-support criterion

Define the multiplicative distortion from the target ray q to the package support cone by

$$R^*(A; q) = \min_{w \geq 0,\, t \geq 0} t \quad \text{s.t.} \quad q \leq Aw \leq tq. \tag{1}$$

We write $R^*(A) \equiv R^*(A; \mathbf{1})$ for the neutral audit used in our released-system tables.

Theorem 1 (target realizability in the support cone). For any $q > 0$, $R^*(A; q) = 1$ if and only if $q \in \operatorname{cone}(A) = \{Aw : w \geq 0\}$. Otherwise $R^*(A; q) > 1$ is the tight minimum componentwise multiplicative distortion from q achievable by package-only mass.

Proof. The constraints force $t \geq 1$. At $t = 1$ they reduce exactly to $Aw = q$, which is equivalent to $q \in \operatorname{cone}(A)$. For any feasible w, the smallest admissible t is $\max_i (Aw)_i/q_i$ after scaling w so that $\min_i (Aw)_i/q_i = 1$; minimizing over w gives the tight distortion. $\square$

This separates STRI from a uniformity assumption: $q = \mathbf{1}$ is the neutral target, not the definition of representation invariance. It also makes exact clone/split invariance immediate: duplicating a support column does not change $\operatorname{cone}(A)$, so it cannot change $R^*(A; q)$ for any fixed q.

Proposition 3 (dual certificate). The dual of Eq. (1) is

$$D^*(A; q) = \max_{\alpha, \beta \geq 0} q^\top \alpha \quad \text{s.t.} \quad A^\top(\beta - \alpha) \geq 0, \quad q^\top \beta \leq 1. \tag{2}$$

Strong duality gives $D^*(A; q) = R^*(A; q)$. Thus every dual-feasible pair is a certified lower bound, while an optimum is complete even when no local witness template is available. On every audited neutral-target regime the numerical primal–dual gap is below 10^{-8}.

3.2 Human-readable factor-2 witness

Theorem 2 (global-singleton overlap lower bound). Suppose packages a, b each have a context supported by exactly that one package over the full frozen support universe, and another context

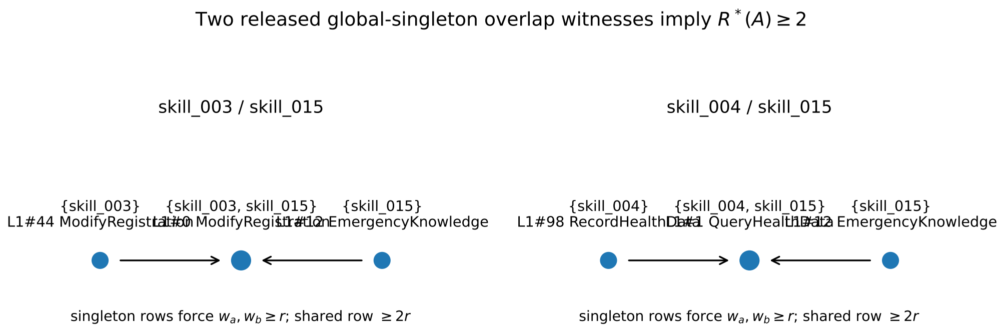

Figure 2: Two released global-singleton overlap witnesses. In each case, retaining positive exposure on the two singleton regions forces at least twice that minimum exposure on the shared region.

is supported by both a and b, possibly with additional packages. If both singleton contexts retain positive exposure, then every nonnegative package mass has max/min exposure ratio at least two on the focal rows, and therefore $R^*(A) \geq 2$ for the full support matrix.

Proof. Let $r > 0$ be the minimum exposure among the two singleton contexts and the shared context. Since the first singleton support signature is exactly $\{a\}$, its exposure is $w_a \geq r$; likewise $w_b \geq r$. The shared context contains both a and b, and additional packages have nonnegative weight, so its exposure is at least $w_a + w_b \geq 2r$. Thus every w has focal max/min ratio at least two. For the same w, the full-matrix maximum is no smaller than the focal maximum and the full-matrix minimum is no larger than the focal minimum, so the full-matrix ratio is also at least two. Minimizing over w gives $R^*(A) \geq 2$. $\square$

For the neutral target $q = \mathbf{1}$, Theorem 2 is also a sparse dual certificate: put $\alpha = 1$ on the two singleton rows, $\beta = 1$ on their shared row, and zero elsewhere. Dual feasibility follows from the support pattern and the objective is two. In released Skill-SP Level-1, the LP dual recovers such a sparse optimum and the primal attains $R^* = 2$.

3.3 STRI-CERT AND DEPLOYABLE VARIANTS

STRI-Cert is a diagnostic protocol, not a learned predictor or a new LP solver: freeze independently defined support and target q; solve Eq. (1) and its dual; report target-realizable iff $R^* = 1$, otherwise the tight residual; expose a sparse witness when available; and fail closed when support, target provenance, or optimization is invalid. Arbitrary nonnegative global package reweighting already receives the complete frozen support matrix, making text deduplication, clone removal, uniform routing, and scalar retuning weaker same-information reductions.

Two extensions make further assumptions explicit. For graded support with independently justified $0 \leq L \leq A \leq U$, the exact box-robust audit replaces Eq. (1) by $Lw \geq q$ and $Uw \leq tq$; estimating valid bounds remains a separate identification problem. Deployment constraints can likewise be audited rather than ignored: adding $w_j \leq \rho \sum_k w_k$ for every package defines a max-share constrained $R^*_\rho(A; q) \geq R^*(A; q)$. Both remain linear programs.

4 RELEASED-SYSTEM EVIDENCE

4.1 SKILL-SP TOOL-CALL SUPPORT AND RELEASED-CONTROLLER CONSEQUENCE

On API-Bank Level-1, 314 rows are accepted by at least one released Skill-SP validator and 183 have multiple memberships. The neutral-target STRI-Cert gives $R^* = 2.0$.

This residual is not only a hypothetical optimum. In the released question-generation path, package-local quality/exploration weights drive identity-weighted sampling before the task exists. At initial-

Table 1: Two-stage audit across released systems. We test observable identity sensitivity first and run STRI-Cert only when independent support truth exists; absence of a support contract is not repaired with candidate-defined labels.

System	Identity perturbation	Independent A?	Audit role
Skill-SP	Pristine sampler gives $2\times$ eligibility; sampler-input clone changes class mass $1/15 \rightarrow 1/8$, while quotient conservation restores it exactly	Yes: released validators	Controller violation + exact clone repair + geometry certificate
SkillRL	Fresh-ID exact clones: 11/12 change semantic retrieval over 223 tasks	No explicit support contract	Phenotype + qualified downstream bridge; no R^*

Table 2: Matched baseline ladder on the Skill-SP tool-call support matrix. Every baseline uses the same frozen released support information; the ladder deliberately strengthens from name-level duplicate removal to the exact best context-independent package reweighting.

Baseline / reduction	Outcome	Evidence
Released name+description Jaccard duplicate filter	Does not absorb	Misses all 5/5 observed specific–generic support-overlap pairs; similarity 0.1429–0.20 is below the released 0.33 threshold.
Exact semantic-clone deduplication	Does not absorb partial overlap	The positive regime uses distinct specific/generic packages with different unique support regions.
Minimum-cardinality whole-package pruning preserving exact support	Partial reduction only	Active packages drop from 10 to 7, but 71 overlap rows remain while exact support is preserved.
Arbitrary nonnegative global package weights	Exact neutral-target residual	The exact optimum is $R^*(A; \mathbf{1}) = 2$.
Global-singleton overlap witness	Sparse dual lower bound	Two independent released witnesses imply $R^*(A; \mathbf{1}) \geq 2$, and the exact LP attains 2.

ization, all 15 tool-call packages have identical zero-attempt statistics, so the released weighting function gives each ID probability $1/15$; a double-supported Level-1 row then receives eligibility opportunity $2/15$ versus $1/15$ for a singleton. The source paper separately reports that replacing dynamic routing by uniform routing lowers Qwen3-4B overall tool-call accuracy from 66.7 to 64.8 (-1.9) (Huang et al., 2026), establishing allocation as a performance-relevant control surface; we do not attribute that gap to STRI distortion.

We next isolate representation sensitivity of the *sampler mapping* with a counterfactual reparameterization: one Level-1-active package at a time is copied with identical content, support, statistics, and sampler-relevant state but a fresh ID, and the author's own `sampling_weights` function is rerun on the augmented library. This intervention is applied directly at the sampler input; it is *not* a claim that the author's induction-time admission path would store a literal exact text duplicate. Indeed, the released name+description Jaccard filter detects each literal clone at similarity 1.0, above its 0.33 rejection threshold. Conditional on the representation intervention, however, the released weighting rule changes the cloned implementation family's mass from $1/15$ to $2/16 = 1/8$ (+87.5%) and every untouched package from $1/15$ to $1/16$ (-6.25%). For each of the six Level-1-active targets, the induced eligibility-exposure ratio changes from 2 to 3 and profile TV is 0.006–0.092. The author's questioner message builder does not expose package ID, so each same-content clone produces exactly the same message string as its source. There are 15 distinct initial message classes: the cloned class changes from $1/15$ to $2/16$, while each of the other 14 changes from $1/15$ to $1/16$; prompt-mixture TV is therefore $7/120 = 0.0583$. In the quotient-conserved control implied by Proposition 1, the original class mass $1/15$ is split $1/30 + 1/30$ while all other class masses stay fixed, recovering both prompt-mixture TV and all six exposure profiles exactly (TV $= 0$, ratio $= 2$). This repair assumes the exact semantic class is known. Prompt-mixture TV is a distribution over actual model inputs; the generated-task distribution additionally depends on the stochastic questioner conditional on those prompts, so we do not equate the two or infer utility.

Table 3: Neutral-target ($q = \mathbf{1}$) mechanism boundary across real first-party support regimes. High overlap is neither necessary nor sufficient; the discriminator is target realizability in the support cone.

Regime	Covered	Multi	Overlap	R^*	Certificate
API-Bank Level-1 full	314	183	58.3%	2.0	Residual
Level-1 calibration tools	47	33	70.2%	2.0	Residual
Level-1 tool-disjoint heldout	52	38	73.1%	2.0	Residual
API-Bank Level-3	34	0	0.0%	1.0	Equalizable
Logical compiler validation	128	127	99.2%	1.0	Equalizable

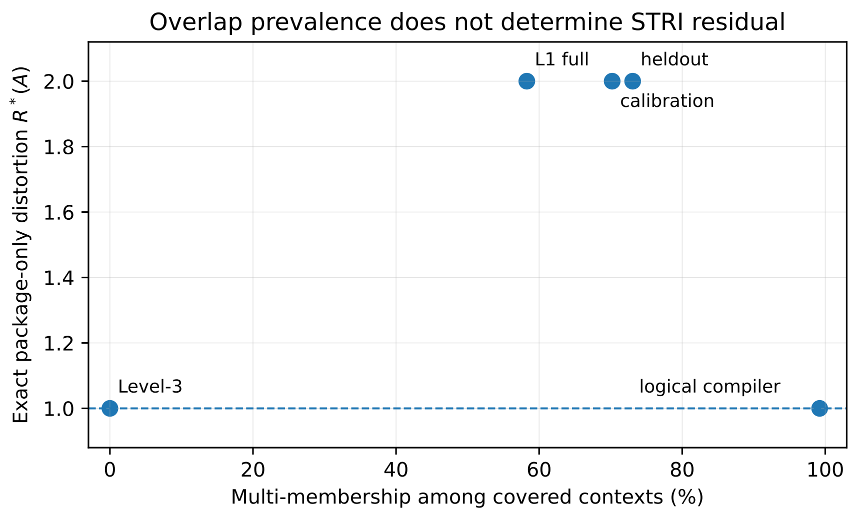

Figure 3: Overlap prevalence does not determine STRI residual. The three Level-1 regimes independently have $R^*(A) = 2$, while Level-3 is equalizable with disjoint support and the logical compiler domain is equalizable despite 127/128 multi-membership tasks.

To test whether the structural residual is an artifact of fitting package weights to the complete table, we use a frozen tool-disjoint split. Calibration has 47 covered tasks, 33 multi-membership rows, and $R^* = 2$; heldout has 52 covered tasks, 38 multi-membership rows, and again $R^* = 2$. Both independently contain the two globally valid factor-2 witness structures.

The witness rows are directly inspectable. For `skill_003`/`skill_015`, L1#44 and L1#12 are singleton regions while L1#0 is shared. For `skill_004`/`skill_015`, L1#98 and L1#12 are singleton regions while L1#1 is shared. In each case Theorem 2 yields the same lower bound attained by the exact LP.

4.2 Level-3: disjoint-support negative control

Level-3 provides a natural released negative regime. Thirty-four rows are covered, every covered row has exactly one skill membership, and $R^* = 1$. Equal weights on the active packages equalize every covered task. STRI-Cert therefore does not mechanically label every released taxonomy problematic.

4.3 Logical compiler domain: high overlap but no residual

A stronger negative test asks whether high overlap alone triggers the certificate. Skill-SP's logical-reasoning release contains eight skill packages with programmatic compiler specifications. The author code can generate a zebra puzzle from a compiler, verify it with a fixed CSP solver, and check whether a generated sample satisfies each package's released compiler-alignment contract.

We freeze the author's default compiler-validation sizes $(3\times3, 4\times4, 5\times5, 6\times6)$ and seeds $0, 1, 2, 3$ for each of eight released skills, giving 128 source units. No language model is used. All 128/128 units compile and pass the author's puzzle contract and uniqueness verifier; every task is then cross-evaluated against all eight released support specifications.

The support matrix is extremely overlapping: 127/128 tasks have more than one supported skill, with 38 distinct support patterns. Yet $R^* = 1$. The exact LP places all mass on `skill_008`, whose released alignment contract supports every generated unit. This is a valid but concentrated equalizer. Under the practical max-share constraint above, $R^*_{0.75} = 4/3$, $R^*_{0.5} = 2$, and $R^*_{0.25} = 4$. Thus the umbrella is not silently treated as universally benign: it identifies the precise feasible-set assumption under which this high-overlap regime is equalizable.

Thus high overlap is not sufficient for a residual: neutral-target equalizability depends on support geometry and the controller's declared feasible package-mass set.

4.4 SkillRL: Independent phenotype, bounded role

SkillRL confirms identity sensitivity in a different control surface: fresh-ID exact-content clones are admitted for 12/12 tested skills, 11/12 change semantic retrieval over 223 ALFWorld descriptions, and 5/12 reduce unique general-skill content under `top_k=6`. It does not expose independent support truth for R^*, so we do not retrofit a candidate-defined A; a qualified endpoint bridge found no success disagreement beyond placebo and is not a population-level no-effect theorem.

4.5 AutoSkill: Dynamic behavioral propagation

AutoSkill (Yang et al., 2026) exposes a first-party `sdk.search` seam. On a frozen five-skill P19 library with `top_k=5`, the original representation retrieves five semantic skills including a post-checkout reset/clean skill. Splitting the highest-ranked Makefile skill into four exact-semantic IDs makes the identity-level top-k contain four copies plus one Python-cleanup skill, deterministically crowding out the post-checkout skill. An ID-renaming placebo preserves the original semantic set, while exact-semantic quotienting before the same five-slot budget restores it.

With the same frozen prompt, Doubao executor, fresh Codex container, and zero judges, the mechanically defined destructive post-checkout behavior occurs in 6/6 original, 0/6 split, 3/3 ID-placebo, and 3/3 quotient-control runs (one-sided Fisher $p = 0.00108$). Holding the split representation and a five-slot/three-class injection fixed, adding back the crowded-out post-checkout skill restores the behavior in 3/3 fresh runs, while a matched cleanup add-back yields 0/3 (exact Fisher $= 1/20$). A sequence-aware parser handles direct or staged hook construction and preserves all 18 Stage-3 labels. This isolates the mediator on one P19 substrate, not AutoSkill utility or general safety.

4.6 Matched baselines, representation ablations, and failure analysis

Table 2 is an explicit same-information baseline ladder, not a collection of unrelated heuristics. The first three baselines may inspect the same complete released support object but can only remove duplicate package columns or whole packages. Minimum exact-coverage pruning is already strong: it removes 61.2% of the observed Level-1 overlap (183 to 71 multi-membership rows). Yet the residual survives. The strongest context-independent baseline then receives arbitrary nonnegative package weights and the full support matrix; its exact optimum remains $R^* = 2$ on the full, calibration, and heldout Level-1 regimes. This is the point at which a simpler package-level explanation is exhausted.

We next ablate the representation itself (Figure 4A). Each active Level-1 package is cloned and split while preserving its support column. Exact column duplication leaves the entire nonnegative support cone unchanged, hence preserves $R^*(A; q)$ for *every* fixed representation-independent target q even though the released identity-indexed sampler above changes. Before quotienting, cloning `skill_003`, `skill_004`, or `skill_015` changes the two-witness diagnostic to 1, 1, and 0 witnesses. Exact-support quotienting removes that syntactic nuisance and recovers every original support row and both witnesses; bijective ID renaming is likewise invariant. Thus quotienting aligns the interpretable witness layer with the exact cone-level certificate.

Finally, we locate both witness and support sensitivity. Across 49 Level-1 tools, 4 have a closed-form witness, 10 have overlap without that template, 20 are disjoint/singleton-only, and 15 have no support (Figure 4C). The exact per-tool LP resolves all 10 overlap-without-witness cases as $R^* = 1$; all four

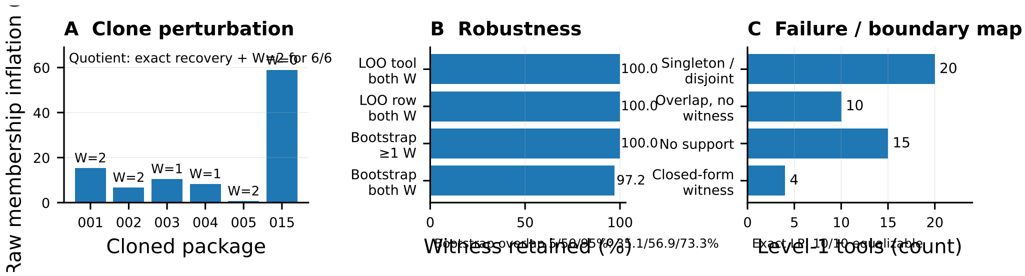

Figure 4: Representation-ablation and robustness evidence. **A:** clone-induced raw multiplicity/witness changes vanish under exact-support quotienting. **B:** factor-2 witnesses survive deletion tests and 500 tool resamples. **C:** the simple witness is only a template; exact LP subsequently resolves the ten overlap/no-witness tools as equalizable.

tool-local residuals are caught by the sparse witness. This does not make Theorem 2 complete in general—Eq. (2) is the authority when the obstruction is nonlocal.

The aggregate positive result is substantially more stable than mere witness resampling suggests. Exhaustively flipping every single zero support cell to one gives 1,387 perturbed Level-1 matrices, all still residual ($R^* \in [2, 3]$). Removing each support edge that leaves its row covered gives 366 more matrices, all with $R^* = 2$; 131 deletions that would uncover a row fail closed. The negative boundaries reveal the opposite asymmetry: every one of 102 possible Level-3 support additions changes $R^* = 1$ to 2, while the logical domain survives all 428 additions but only 468/595 non-uncovering deletions; 127/595 break equalizability. These are exact one-cell finite-snapshot stress tests, not a claim that a learned support estimator is calibrated or that arbitrary multi-cell noise is harmless. The earlier leave-one-tool/row and 500 fixed-seed tool-bootstrap tests remain consistent: both Level-1 witnesses survive every deletion; at least one survives 100% and both 97.2% of bootstrap replicates.

5 Relation to Existing Work

Skill redundancy, governance, and safety. Clone detection and lifecycle governance already detect, suppress, merge, or update redundant skills (Zhu et al., 2026; Liu et al., 2026); safety studies show that skill content can propagate failures (Mao et al., 2026; Dong et al., 2026). STRI claims neither clone detection nor harmful content: it asks whether unchanged semantics receive different control solely from representation, and its partial-overlap residual survives exact-clone reasoning.

Evolving controllers. ERSkill, Skill-SP, SkillRL, SHAPER, and related systems learn routers or co-evolve skill libraries (Chen et al., 2026; Huang et al., 2026; Xia et al., 2026; Wang et al., 2026). They motivate STRI because package IDs become endogenous control objects, but do not characterize invariance under semantics-preserving reparameterization.

Classical optimization boundary. Skill compression and structured bandits address reusable structure or information sharing (Bai et al., 2026; Slivkins, 2011; Sen et al., 2018). Our LP is adjacent to fractional covering, matrix scaling, max–min fairness, and discrepancy (Lovász, 1975; Sinkhorn & Knopp, 1967; Radunović & Le Boudec, 2007; Beck & Fiala, 1981); we claim none of those algorithms. Their objectives are covering cost, prescribed marginals, fairness, or signed balancing, whereas STRI tests whether semantic control is invariant to an endogenous taxonomy representation.

6 Discussion and Claim Boundary

STRI-Cert audits target realizability under declared package mass. We claim released representation sensitivity, exact support-cone realizability, and one bounded AutoSkill P19 behavior chain with placebo, quotient, and mediator controls; we **do not** claim task utility, longitudinal regret, system-wide safety, LP novelty, or a broadly validated repair. Exact repair and R^* need clone/support/target truth; dynamic evidence is one P19 substrate/executor; neutral $q = \mathbf{1}$ is only the no-priority audit.

AI USE STATEMENT

In this work, generative AI tools assisted with conceptual framing and hypothesis refinement; formulation and critique of mathematical claims and proof drafts; experiment and falsifier design; implementation and review of analysis, experiment-control, and manuscript-quality-assurance code; interpretation of intermediate results; primary-source literature triage; and preparation or editing of manuscript text, figures, tables, and references. Generative models were also used as the task proposer in a preregistered Qwen3-8B dynamic pilot; that pilot failed its frozen author-native qualification gate, remains documented in the supplement, and is excluded from scientific claims about dynamic STRI. Load-bearing numerical claims in the submitted narrow paper are bound to released first-party artifacts or deterministic/programmatic analysis outputs. The support certificate, theorem edge cases, transition logic, and paper-level numerical/citation constraints were exercised by executable tests; related-work boundaries and bibliographic metadata were checked against primary or official sources. AI-assisted outputs were not treated as independent scientific authority. We take responsibility for the final content of this work, including text, claims, code, and artifacts produced with the aid of generative AI.

REPRODUCIBILITY STATEMENT

The main paper states the exact-refinement quotient characterization, semantic-first design construction, target-conditioned primal/dual programs, closed-form witness, box-robust extension, and constrained-controller audit used for the load-bearing claims. The Skill-SP controller audit is bound to author commit `bb693c89fee66e1f824d6a777759a49b7a295a83`, released `library.py` SHA-256 `8441de35...e2756e0`, question-generation SHA-256 `1b4cb67f...82d739b`, initial package-bundle SHA-256 `d4c4350a...68b6e7`, and tool-call membership SHA-256 `b73d8637...6f2e17d4`. The supplement packages our controller-audit/certificate code, tests, frozen receipts and matrices, but intentionally does not redistribute the third-party Skill-SP repository; its README gives the exact author commit required for end-to-end rerun. LPs use SciPy HiGHS; certificate classification uses tolerance 10^{-8}. Executable tests cover the target-conditioned dual, target-ray scaling, semantic-first target realization under arbitrary overlap, quotient clone invariance, sampler-weight recomputation, literal-duplicate rejection, exact questioner-message equality, the $7/120$ prompt-mixture shift, quotient restoration to zero, one-cell support perturbations, and per-tool LP results. All dynamic thresholds were frozen before outcomes. The failed Qwen3-8B bank remains a transparent non-result; the qualified SkillRL bridge separately records 18/24 competence calibration, 24 paired A/B/C/D units, exact A/D coupling, and the statistical-resolution boundary that prevents upgrading a realization STOP into a population-level no-effect claim.

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